

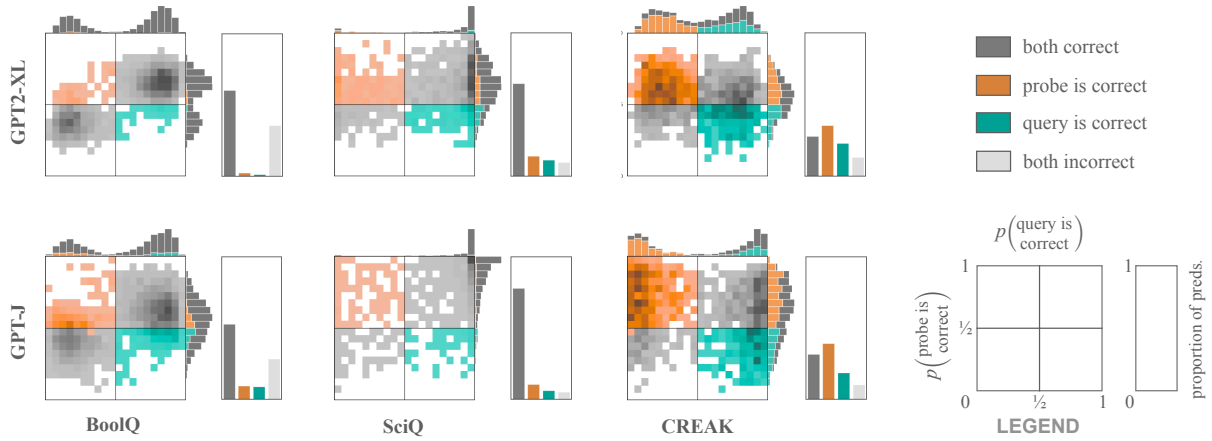


Figure 3: Distribution of query and probe predictions. “Deception”-like results (orange) do not feature particularly prominently compared to other outcomes. Note that heatmaps (left) use the same shade of gray for “both correct” and “both incorrect”, even though they are distinguished in histograms (right) to enable direct comparison.

than 20% on CREAK).

Another striking difference between probes and queries is visualized in Fig. 2, which plots the calibration of predictions from GPT-J (e.g. when a model predicts an answer with 70% probability, is it right 70% of the time?). Probes are well calibrated, while queries are reasonably well calibrated on SciQ but poorly calibrated on other datasets. This aligns with the finding by Mielke et al. (2022) that it is possible to obtain calibrated prediction of model *errors* (without assigning probabilities to specific answers) using a similar probe.

It is important to emphasize that the main goal of this work is to understand representational correlates of truthfulness, not to build more accurate models. Indeed, the superior calibration and accuracy of probes over queries might not seem surprising given that probes are trained many-shot while queries use the model zero-shot. To contextualize these results, we also finetuned GPT2-XL on true question/answer pairs and found it performed better overall than probing the pretrained model. Probe predictions exhibited 1.7% better accuracy than fine-tuned model queries on BoolQ. However, querying the fine-tuned model had 6.9% and 4.0% better accuracy on SciQ and CREAK.

4 Most disagreements result from confabulation and heterogeneity

Next, we study the joint distribution of query and probe predictions. Results are visualized in Fig. 3. GPT2-XL and GPT-J exhibit a similar pattern of errors on each dataset, but that distribution varies

substantially between datasets. In BoolQ and SciQ, the query is correct and the probe is incorrect nearly as often as the opposite case; on both datasets, examples of highly confident but contradictory predictions by probes and queries are rare.

Fig. 1D shows the fraction of examples in each of the nine categories depicted in Fig. 1C. Only in CREAK do we observe a significant number of instances of “deception”; however, we also observe many instances of probe errors. In all other cases (even SciQ, where probes are substantially more accurate than queries), query–probe mismatches predominantly represent other types of disagreement.

5 Queries and probes are complementary

Another observation in Fig. 1 is that almost all datasets exhibit a substantial portion of heterogeneity: there are large subsets of the data in which probes have low confidence, but queries assign high probability to the correct answer (and vice-versa).

We can exploit this fact by constructing an *ensemble* model of the form:

$$p_{\text{ensemble}}(a | q) = \lambda \cdot p_{\text{probe}}(a | q) + (1 - \lambda) \cdot p_{\text{query}}(a | q) \quad (1)$$

We select λ using 500 validation examples from each dataset, and evaluate accuracy on the test split. Results are shown at the bottom of Table 1; in 4/6 cases, this ensemble model is able to improve over the probe. While improvements on BoolQ and CREAK are small, on SciQ they are substantial—nearly as large as the improvement of the probe

over the base LM. These results underscore the fact that, even when probes significantly outperform queries, query errors are not the source of all mismatches, and heterogeneity in probing and querying pathways can be exploited.

6 Conclusion

We studied mismatches between language models’ factual assertions and probes trained on their internal states, and showed that these mismatches reflect a diverse set of situations including confabulation, heterogeneity, and a small number of instances of deception. In current models, disagreements between internal and external representations of truthfulness appear predominantly attributable to different prediction pathways, rather than a latent intent to produce incorrect output. A variety of other model interventions are known to *decrease* truthfulness, including carefully designed prompts (Lin et al., 2022), and future models may exhibit more complex relationships between internal representations and generated text (especially for open-ended generation). Even in these cases, we expect the taxonomy in Fig. 1 to remain useful: not all mismatches between model and probe behavior involve deception, and not all model behaviors are (currently) reducible to easily decodable properties of their internal states.

7 Limitations

As seen in Fig. 1, there is significant heterogeneity in the distribution of disagreement types across datasets. These specific findings may thus not predict the distribution of disagreements in future datasets. As noted in Section 2, our experiments use only a single prompt template for each experiment; we believe it is likely (especially in CREAK) that better prompts exist that would substantially alter the distribution of disagreements—our goal in this work has been to establish a taxonomy of errors for future models. Finally, we have presented only results on linear probes. The success of ensembling methods means that some information must be encoded non-linearly in model representations.

8 Ethical Considerations

Our work is motivated by ethical concerns raised in past work (e.g., Askell et al., 2021; Evans et al., 2021) that LMs might (perhaps unintentionally) mislead users. The techniques presented here might be used to improve model truthfulness or detect

errors. However, better understanding of model-internal representations of factuality and truthfulness might enable system developers to steer models toward undesirable or harmful behaviors.

Finally, while we have presented techniques for slightly improving the accuracy of models on question answering tasks, models continue to make a significant number of errors, and are not suitable for deployment in applications where factuality and reliability are required.

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